

CS215 Assignment-1

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1 Solution Q1

Let $w_{P,k}$ represent the event that player P gets a prime number on k^{th} throw, and hence gets a win.

$$w_{P,k} = P(2, 3, 5) = \frac{3}{6} = \frac{1}{2} \text{ for both players A and B}$$

Let X_r and Y_r denote the random variables representing no of wins by A and B respectively in first r rolls. Now,

$$P(X_{n+1} > Y_n) = P(X_n > Y_n) \cdot (1) + P(X_n = Y_n) \cdot P(w_{A,n+1})$$

as either A has more wins in first n rolls itself, or A gets the last win in $(n+1)^{\text{th}}$ roll.

Now,

$$\begin{aligned} P(X_n = Y_n = r) &= P(X_n = r) \cdot P(Y_n = r) = P(X_n = r)^2 \\ &= \left(\binom{n}{r} w^r (1-w)^{n-r} \right)^2 = \left(\binom{n}{r} \left(\frac{1}{2} \right)^r \left(\frac{1}{2} \right)^{n-r} \right)^2 \\ &= \frac{1}{2^{2n}} \cdot \binom{n}{r}^2 \end{aligned}$$

$$\begin{aligned} P(X_n = Y_n) &= \sum_{r=0}^n \frac{1}{2^{2n}} \binom{n}{r}^2 \\ &= \frac{1}{2^{2n}} \cdot \binom{2n}{n} \end{aligned}$$

By symmetry, $P(X_n > Y_n) = P(X_n < Y_n)$

$$1 = P(X_n > Y_n) + P(X_n = Y_n) + P(X_n < Y_n)$$

$$\Rightarrow P(X_n > Y_n) = \frac{1 - P(X_n = Y_n)}{2}$$

$$\Rightarrow P(X_n > Y_n) = \frac{1}{2^{2n+1}} \cdot \left(2^{2n} - \binom{2n}{n} \right)$$

So, finally

$$\begin{aligned} P(X_{n+1} > Y_n) &= \frac{1}{2^{2n+1}} \cdot \left(2^{2n} - \binom{2n}{n} \right) + \frac{1}{2^{2n}} \cdot \binom{2n}{n} \cdot \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

2 Solution Q5

Assume that the IDs of the traders are randomly distributed.

Let the probability of me winning if I choose my position to be r be $F(r)$.

$$F(r) = P(\text{I win} \mid \text{I am at position } r) = P(\text{Traders at } 1, 2, \dots, r-1 \text{ lose}) \times P(\text{I win at position } r)$$

$$F(r) = \begin{cases} 0 & \text{if } r = 1, \\ \frac{200}{200} \cdot \frac{199}{200} \cdot \dots \cdot \frac{200-(r-2)}{200} \cdot \frac{r-1}{200} & \text{if } 2 \leq r \leq 200, \\ 0 & \text{if } r > 200. \end{cases}$$

Now, suppose my winning probability is maximum at $r = r_0$.

Then, $F(r) > F(r-1)$ and $F(r) > F(r+1)$.

$$\begin{aligned} \frac{200}{200} \dots \frac{203-r}{200} \cdot \frac{202-r}{200} \cdot \frac{r-1}{200} &> \frac{200}{200} \dots \frac{203-r}{200} \cdot \frac{r-2}{200} \\ \frac{200}{200} \dots \frac{202-r}{200} \cdot \frac{r-1}{200} &> \frac{200}{200} \dots \frac{202-r}{200} \cdot \frac{201-r}{200} \cdot \frac{r}{200} \end{aligned}$$

$$\begin{aligned} \frac{202-r}{200} \cdot (r-1) &> (r-2) & \frac{201-r}{200} \cdot r &< r-1 \\ \Rightarrow r^2 - 203r + 202 &< 200r - 400 & \Rightarrow r^2 - 201r + 200r - 200 &> 0 \\ \Rightarrow r^2 - 3r - 198 &< 0 & \Rightarrow r^2 - r - 200 &> 0 \\ \Rightarrow r \in (-12.65, 15.65) & & \Rightarrow r \in (-\infty, -13.65) \cup (14.65, \infty) \end{aligned}$$

Hence, by taking the intersection of the above intervals, I conclude that the probability of me winning is maximum at $r = 15$.